

STAT 482 final project (Group 8)

Linking Equities and Crypto: Return, Volatility, and Diversification

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Introduction

Motivation

- Over the last decade, crypto has evolved from a niche product to a major asset class held by both retail and institutional investors.
- At the same time, U.S. equity indices such as the S&P 500 and Nasdaq remain the core building blocks of most diversified portfolios.
- Practitioners face two key questions:
 - *Does crypto truly diversify an equity portfolio?*
 - *Or does it simply behave like a high-beta extension of equity risk?*
- Since crypto markets are relatively young, empirical evidence on how they interact with traditional assets is still limited.
- Our goal is to provide data-driven evidence on the link between equities and crypto using standard time-series tools.

Research Objectives

Primary Objective:

Quantify and interpret the relationship between Major U.S equity indices and leading crypto assets in terms of:

- **How closely do crypto and equity markets move together, and if that relationship changes over time**
- **If changes in one market can help predict or influence the other market**
- **Interpret the two former testings' results in a real world portfolio**

Dependence

Estimate the magnitude of return dependence and evaluate its sensitivity across time and market regimes

Dynamics

Test crypto-equity market's predictive influence on each other using Granger causality and evaluate cross-market shock

Implications:

Given results from previous testings, evaluate real-world implications through possible gains in portfolio efficiency

Data & Analysis

Data

Assets:

- Equity indices:
 - S&P 500 Index (SP500)
 - Nasdaq Composite Index (Nasdaq)
- Crypto assets:
 - Bitcoin (BTC)
 - Ethereum (ETH)
 - Solana (SOL)

Sample and data source:

- Source: Yahoo Finance
- Sample period: 2018-01-01 to the most recent available date
- Keep only dates where all five assets have valid prices to ensure aligned panels.
- Analyze both:
 - Daily data (main focus), and
 - Weekly data (using last trading day of each week)

Return Construction

From prices to returns, for each asset, we construct log returns:

$$r_t = \log(P_t) - \log(P_{t-1})$$

We construct and obtain:

- log returns from daily closing prices
- Weekly log returns from weekly closing prices

Cleaning and alignment:

- Merge all assets into single price panel and drop any rows with missing values, so each observation includes SP500, Nasdaq, BTC, ETH, and SOL on the same date
- All analyses are performed on cleaned data series, not on raw price levels.

Roadmap

- 1 Descriptive Statistics & Correlation Structure

- 2 Dynamic Return Linkages (VAR & Granger)

- 3 Volatility & Time Correlations (GARCH & DCC)

- 4 Predictive Power & Model Comparison)

- 5 Portfolio & Diversification Effects

- 6 Regime & Tail Dependence Analysis

- 7 Equity → Crypto Volatility Spillovers (Realized Vol & Predictability)

Block 1

1. Descriptive Statistics & Correlation Structure

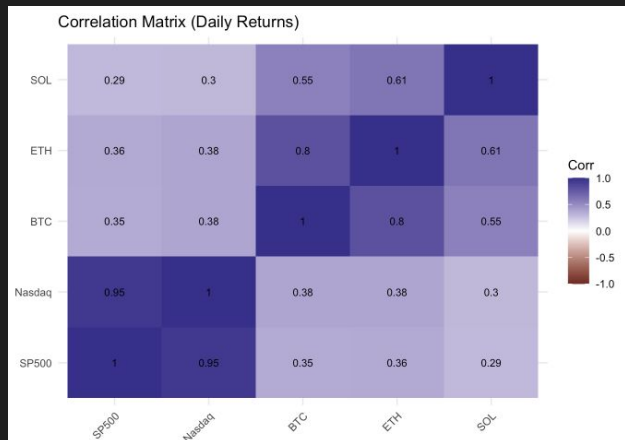
Levels and risk

- **Equities vs. crypto:**
 - SP500/Nasdaq have small daily means (~0.06-0.07%) and low volatility (~1%)
 - Crypto assets have higher average returns but ~3-7x higher volatility.
- **Heavy tails:**
 - Skewness & kurtosis are much larger for crypto
 - returns are more extreme and non-Gaussian, especially for SOL
- **Conclusion:**
 - So far, the combination of higher volatility and heavy tail behavior in crypto simply indicates larger magnitude, which corresponds with the higher risk profile.

##	Observations	NAs	Minimum	Quartile 1	Median	Arithmetic Mean
## SP500	1417	0	-0.0616	-0.0047	0.0010	0.0006
## Nasdaq	1417	0	-0.0615	-0.0064	0.0014	0.0007
## BTC	1417	0	-0.2572	-0.0162	0.0011	0.0018
## ETH	1417	0	-0.3237	-0.0232	0.0015	0.0020
## SOL	1417	0	-0.5496	-0.0359	0.0009	0.0036
##	Geometric Mean	Quartile 3	Maximum	SE Mean	LCL	Mean (0.95)
## SP500	0.0006	0.0069	0.0909	0.0003		1e-04
## Nasdaq	0.0006	0.0092	0.1148	0.0004		0e+00
## BTC	0.0011	0.0200	0.1915	0.0010		-2e-04
## ETH	0.0008	0.0273	0.3435	0.0013		-6e-04
## SOL	0.0005	0.0394	0.5805	0.0021		-5e-04
##	UCL	Mean (0.95)	Variance	Stdev	Skewness	Kurtosis
## SP500	0.0012	0.0001	0.0110	-0.1911	5.5849	
## Nasdaq	0.0015	0.0002	0.0144	-0.0814	3.9967	
## BTC	0.0037	0.0014	0.0373	-0.1853	4.3567	
## ETH	0.0047	0.0025	0.0503	-0.1342	5.7649	
## SOL	0.0076	0.0060	0.0777	0.2677	8.2435	

1. Descriptive Statistics & Correlation Structure

Static correlation structure (daily vs. weekly)

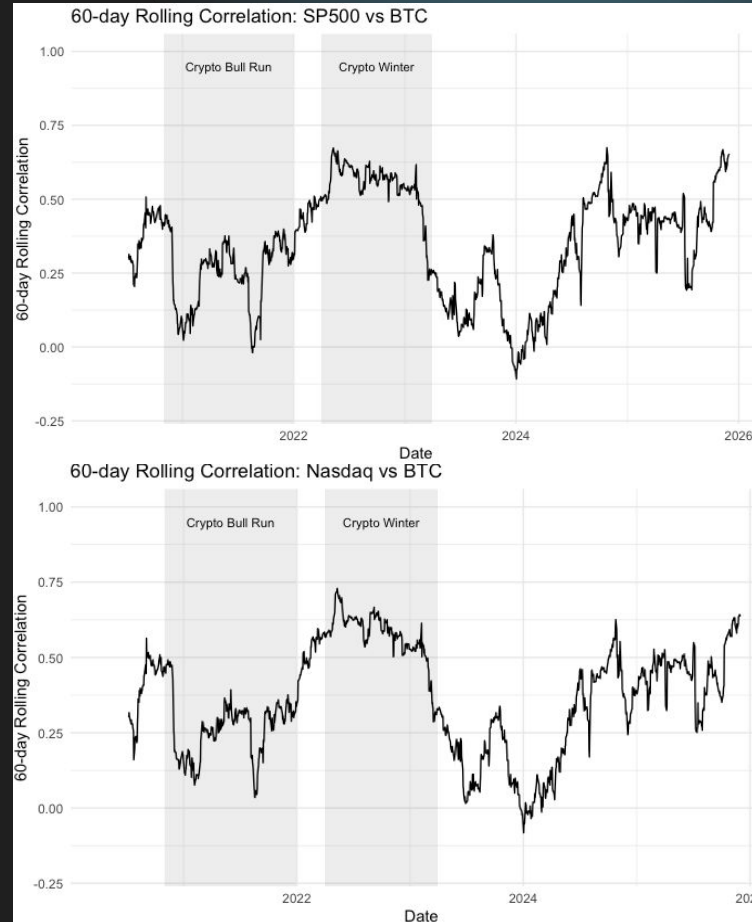


- **Within-equity correlations:**
 - SP500 & Nasdaq almost perfectly correlated ($\rho \approx 0.95$) both daily & weekly
- **Within-crypto correlations:**
 - BTC, ETH, SOL tightly linked ($\rho \approx 0.5$ – 0.8): meaningfully connected to each other but far from 1, which means they share directional tendencies but there is room for individual movement
- **Cross Market correlation:**
 - Equity-to-crypto correlations hover around ($\rho \approx 0.25$ – 0.35), positive but comparatively small.
 - crypto returns co-move with equities, but not enough to suggest these three crypto prices are moved by the same factors as the SP500/Nasdaq.
- **Stable patterns across frequencies:**
 - correlation nearly identical with weekly returns computation, suggesting systemic co-movements

1. Descriptive Statistics & Correlation Structure

Time-varying equity-BTC correlations

- **Correlations are clearly time-varying:**
 - 60-day rolling coefficients move from 0 to roughly 0.7 across time, so the BTC-equity link cannot be summarized by a single constant number.
- **Periods of elevated correlation align with large market moves:**
 - During the 2020–2021 expansion and the 2022 downturn, equity returns mostly sit in the 0.3–0.6 range and sometimes reach 0.6–0.7, meaning BTC tends to move with stocks when equity trends are strong.
 - In mid-2023, equities exhibit low directional momentum and rolling correlations fall toward zero, possibly indicating without strong equity market directions, BTC's return dynamics revert to independent behavior instead of co-movement
 - Across the full window, correlation peaks are on average higher against Nasdaq



Block 2

2. Dynamic Return Linkages: VAR & Granger

VAR setup & overall Granger results

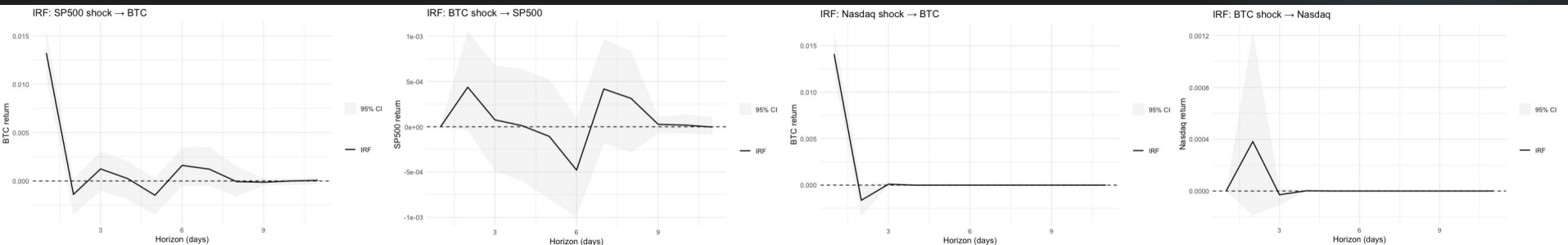
- **Model:**
 - Daily log-returns VAR with SP500, Nasdaq, and BTC
 - Lag order selected by AIC for each specification
 - Hypothesis: “Do equity indices help predict BTC returns, or does BTC help predict equity returns?”
- **Interpretations:**
 - After evaluating for each asset’s history, neither direction shows statistically reliable Granger predictability.
 - Although equities and BTC occasionally align in rolling correlations, the VAR shows that these movements do not extend to systematic forecasting ability.
 - The relationships appear episodic and state-dependent rather than stable transmissions.

- **Results:**

Direction <chr>	p_value <dbl>
SP500 → BTC (joint)	0.5591
Nasdaq → BTC (joint)	0.7776
BTC → (SP500, Nasdaq) (joint)	0.1986

2. Dynamic Return Linkages: VAR & Granger

Impulse responses: equity shocks vs. BTC



- **SP500 shock and BTC:**
 - a positive 1-day SP500 shock is followed by an ~ 1-1.5% BTC increase the next day.
 - Effect decays quickly and is essentially gone after 3-4 days
- **BTC shock and SP500:**
 - The response is very small and mostly inside the 95% band, economically negligible
- **Nasdaq and BTC:**
 - Overall, equity-to-crypto spillovers dominate crypto-to-equity spillovers, but both are short-run and weak
 - Nasdaq influence on BTC looks similar to SP500 influence on BTC but is even smaller in magnitude.
- **Interpretation:**
 - Return spillovers direction is mainly from equities to BTC, not the other way around.
 - Crypto shocks do not seem to impact equity indices: this is consistent with the Granger causality results
 - The short-lived impulse responses imply that co-movement is not due to any systematic predictive impact
 - This might imply that there exist in both markets concurrent reactions to broader market condition

2. Dynamic Return Linkages: VAR & Granger

Robustness: Across Regimes and Coins

- **Regime-specific SP500 to BTC (upper):**
 - full sample: SP500 to BTC $p \approx 0.08$, BTC to SP500 $p \approx 0.32$
 - 2020-2021 and 2022: both directions clearly insignificant ($p > 0.1$)
- **Other coins (lower):**
 - BTC&SOL: SP500/Nasdaq to coin have only weak effects; coin to equities is never significant
 - ETH: SP500/Nasdaq show stronger Granger-causality, but ETH still doesn't predict equities
- **Takeaway:**
 - Across indices, regimes, and coins, at most **modest predictability** from equities to crypto.
 - almost **no evidence that crypto returns Granger-cause** SP500 or Nasdaq.

Regime <chr>	lag_AIC <dbl>	p_SP500_to_BTC <dbl>	p_BTC_to_SP500 <dbl>
Full sample	7	0.0895	0.3096
2020-2021	1	0.6117	0.7312
2022+	1	0.1777	0.1364

Coin <chr>	lag_SP_AIC <dbl>	lag_Nasdaq_AIC <dbl>	p_SP500_to_Coin <dbl>	p_Coin_to_SP500 <dbl>	p_Nasdaq_to_Coin <dbl>	p_Coin_to_Nasdaq <dbl>
BTC	7	1	0.0895	0.3096	0.1705	0.3181
ETH	7	7	0.0013	0.0854	0.0008	0.1041
SOL	1	1	0.2174	0.6845	0.1225	0.6301

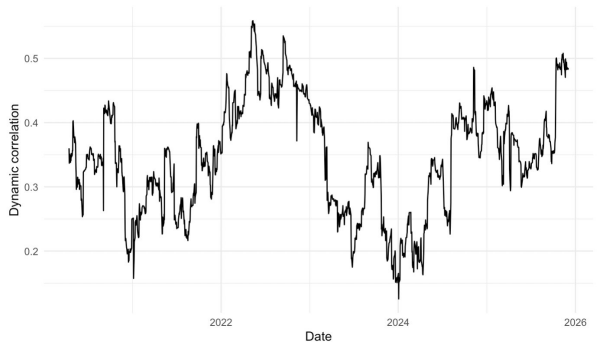
Block 3

3. Time-Varying Correlations: DCC-GARCH

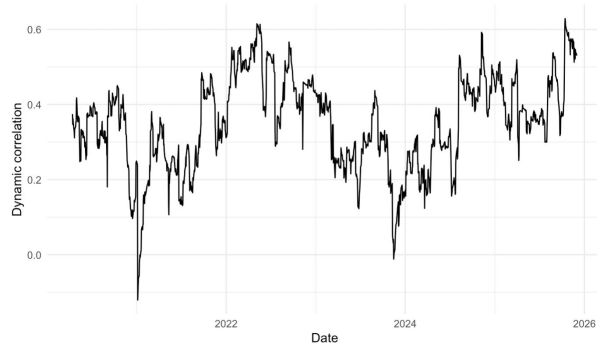
- **Methods:**
 - Bivariate DCC-GARCH on daily log returns of SP500 with BTC, ETH, SOL.
 - Model lets both conditional volatilities **and** conditional correlations evolve over time.
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- **Question::**
 - “How strongly do each crypto co-move with SP500, and how stable is that link?”

3. Time-Varying Correlations: SP500 vs BTC/ETH/SOL

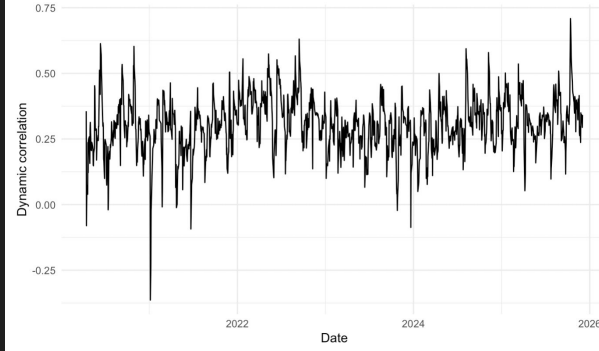
DCC-GARCH Correlation: SP500 – BTC



DCC-GARCH Correlation: SP500 – ETH



DCC-GARCH Correlation: SP500 – SOL



- **Overall levels:**
 - Correlations are **mostly positive** for all three coins.
- **BTC & ETH:**
 - BTC: typically around 0.3–0.4, with spikes near 0.5+.
 - ETH: slightly higher and more volatile, often 0.3–0.5, with brief jumps above 0.6.
- **SOL:**
 - Lower and fluctuates more, averaging around 0.25–0.35, with occasional dips close to zero or negative.
- **Interpretation:**
 - The DCC paths show that equity–crypto links are consistently positive but not strong enough. BTC and ETH usually move in the same direction as the SP500 with correlations in the 0.3–0.4 range, while SOL sits lower, at times even slightly negative.

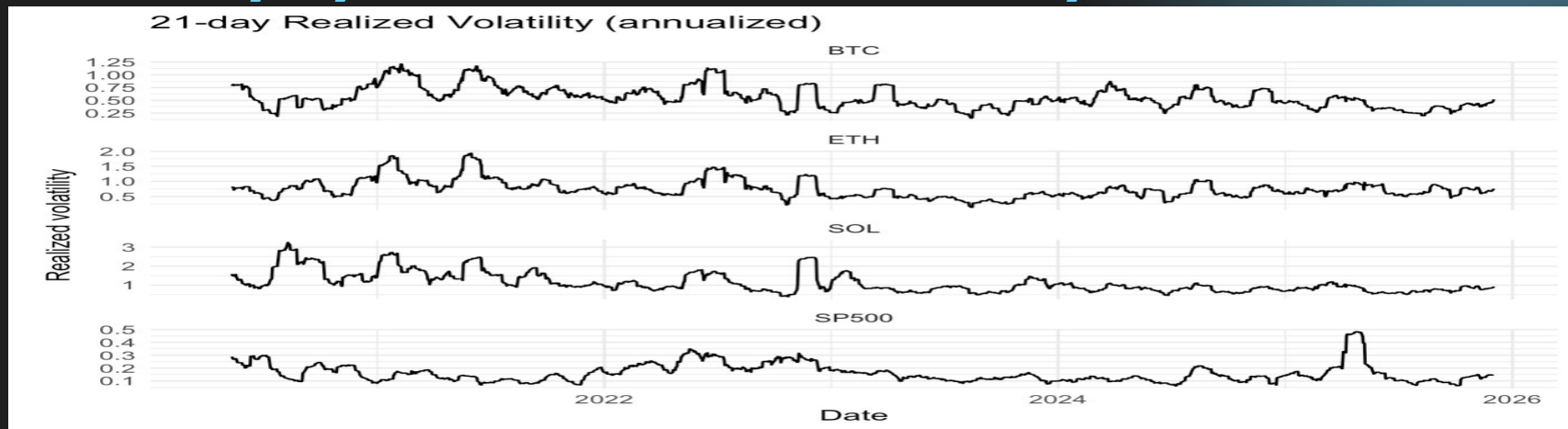
3. Summary of Dynamic Correlations

- **Average co-movement:**
 - Mean correlations: BTC $\approx$ 0.34, ETH $\approx$ 0.35, SOL $\approx$ 0.30.
- **Extremes:**
 - Min: can drop to mildly negative (ETH) and even more for SOL (down to about -0.36).
 - Max: ETH and SOL sometimes spike above 0.6-0.7 in stress episodes.
- **Takeaway:**
 - Across the full sample, the three coins generally point in the same direction as SP500, but the mid-0.3 average correlations leave plenty of room for independent moves.
 - The spread between min and max, from mildly negative to above 0.6-0.7, especially for ETH and SOL, shows that the movement is still highly regime-dependent.
 - Implication: in tranquil or coin-specific phases, crypto can offset some equity risk, but in major equity stress episodes the link tightens, and we will not see that much diversification benefit.

Coin <chr>	mean_corr <dbl>	min_corr <dbl>	max_corr <dbl>
BTC	0.344	0.125	0.559
ETH	0.350	-0.121	0.629
SOL	0.303	-0.364	0.709

Block 4

4. Volatility Dynamics: Realized Volatility Across Markets



- Setup:
 - 21-day rolling realized volatility (annualized) for BTC, ETH, SOL, and SP500.
- Visual Patterns:
 - Crypto vol is consistently several times higher than SP500 vol.
 - All assets show common “risk-on / risk-off” spikes (e.g., 2022 turbulence), but crypto has many extra idiosyncratic jumps.
- Interpretation:
 - The 21-day realized volatility series suggest that all four assets respond to the same broad risk cycles, the big spikes line up around episodes like the 2022 turbulence, but the magnitude of the swings is very different.
 - Crypto volatility typically runs several times above SP500 volatility and is punctuated by frequent, short-lived surges.

4. Volatility Spillovers: VAR & Granger on Realized Vol

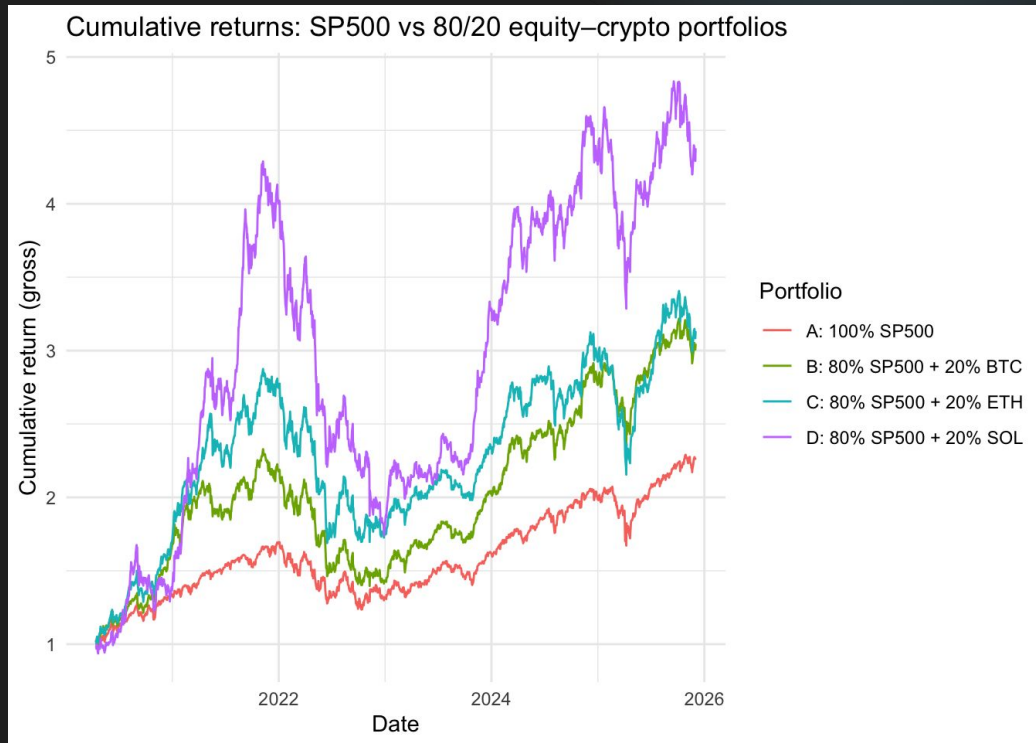
- Model:
 - 2-variable VAR on 21-day realized vol: *SP500vol*, *Coinvol* for BTC, ETH, and SOL.
 - Lag order chosen by AIC for each pair.
- Results (p-values, table below):
 - BTC & SOL: no evidence of SP500 vol to coin vol, and no evidence of coin vol to SP500 vol (all p-values > 0.25).
 - ETH: ETH volatility to SP500 volatility is marginally significant ($p \approx 0.046$); SP500 vol to ETH vol is not ($p \approx 0.30$).
- Takeaway:
 - Volatility spillovers are generally weak in both directions.
 - The single ETH to SP500 result is small and likely not very robust.
 - Even in turbulent periods, SP500 volatility seems mostly driven by equity-side shocks and does not respond to crypto markets.

Coin <chr>	lag_AIC <dbl>	p_SP500vol_to_Coinvol <dbl>	p_Coinvol_to_SP500vol <dbl>
BTC	4	0.6025	0.2725
ETH	5	0.2965	0.0460
SOL	10	0.9139	0.4476

Block 5

5. Diversification Benefit: Equity + Crypto Efficient Frontier

- Return upsides are derived from crypto's higher magnitude returns, not uniform correlation benefits: Crypto disproportionately contributes during high-beta market phases
-
- Performance spread across BTC/ETH/SOL reflects different structural risk: BTC's smaller gain here could suggest a more mature market. On the other hand, ETH and especially SOL show larger separation due to higher realized volatility and asymmetric positive tail events, thus the noticeably higher cumulative return.



5.2: Portfolio Efficiency Analysis

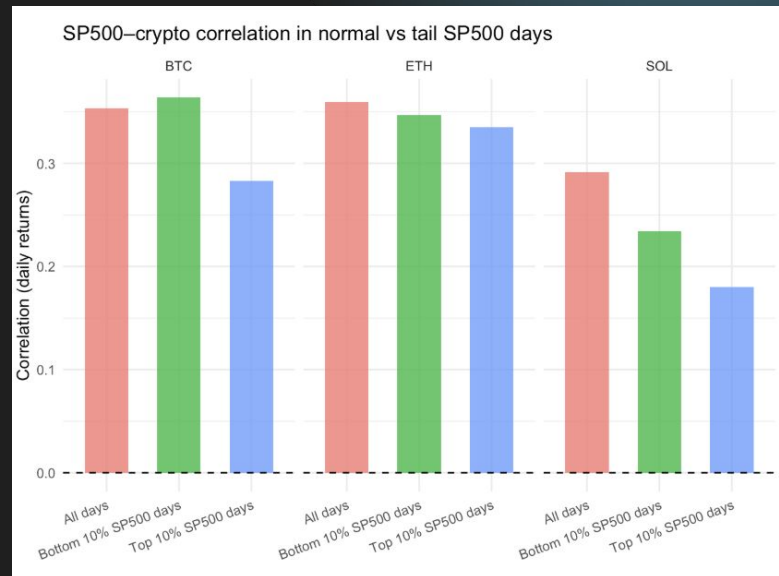
##	Portfolio	Ann_Return	Ann_Vol	Sharpe	MaxDD
## A_SP500_100	A_SP500_100	0.1566	0.1753	0.893	0.2711
## B_80_20_SP500_BTC	B_80_20_SP500_BTC	0.2189	0.2131	1.027	0.4002
## C_80_20_SP500_ETH	C_80_20_SP500_ETH	0.2250	0.2475	0.909	0.4126
## D_80_20_SP500_SOL	D_80_20_SP500_SOL	0.3001	0.3175	0.945	0.5936

- Why sharpe ratio: allows us to understand better on the practical side, tests whether the higher returns from adding crypto compensate for the extra volatility
- Why Maxdrawdown(MaxDD): captures the worst peak-to-trough loss during the sample period. It shows the depth of the worst cumulative decline and represents tail-risk
- Bitcoin: increases Sharpe the most. showed large positive return bursts in periods when equities underperformed. Its high volatility is partly offset by its diversification benefits,
- Eth: diversification benefit relative to SP500 is weaker than BTC. ETH volatility spikes tend to align with broader risk-off episodes
- SOL: raises expected return the most but also drives Sharpe lower. Return profile is explosive but much more speculative and momentum-driven. Huge upside periods and violent corrections. Return jumps, but volatility and drawdown rise even faster, reducing risk-adjusted attractiveness.

Block 6

6: Tail Dependence Equity Up/Down Extremes

```
## # A tibble: 9 × 3
##   Coin Condition      Corr
##   <chr> <chr>      <dbl>
## 1 BTC   All days      0.354
## 2 BTC   Top 10% SP500 days 0.283
## 3 BTC   Bottom 10% SP500 days 0.364
## 4 ETH   All days      0.360
## 5 ETH   Top 10% SP500 days 0.335
## 6 ETH   Bottom 10% SP500 days 0.347
## 7 SOL   All days      0.291
## 8 SOL   Top 10% SP500 days 0.180
## 9 SOL   Bottom 10% SP500 days 0.234
```



- BTC and ETH exhibit asymmetric left-tail dependence: correlations highest on the bottom 10% SP500 days (BTC: 0.364, ETH: 0.347) and lowest on the strongest SP500 days (BTC: 0.283, ETH: 0.335). Both coins move more tightly with the SP500 during equity stress episodes, but partially disconnects during strong equity rallies.
- SOL behaves fundamentally differently in tails: correlations decline in both tails (Top: 0.180, Bottom: 0.234 vs All-days: 0.291), showing that extreme SP500 moves reduce its co-movement.
- Conclusion: higher average correlations in later periods and stronger left-tail co-movement for BTC/ETH imply weaker diversification benefits exactly when equities are under pressure, while SOL offers the strongest tail diversification because its linkage falls during both upside and downside extremes.

Block 7

7. Equity → Crypto Volatility Spillovers

Volatility levels: equities vs. crypto

- Crypto is in a much higher volatility regime: BTC/ETH/SOL typically run at 3-5x the SP500's 21-day volatility
- All assets share several turmoil episodes, especially the late-2021 boom and the 2022-23 crypto downturn, where volatility spikes together
- Post high volatility episodes, the SP500 settles back to a calmer range, while crypto volatility stays elevated.



7. Equity → Crypto Volatility Spillovers

Do equity shocks forecast crypto volatility?

- When we regress 21-day crypto volatility on today's absolute SP500 and Nasdaq returns, the R^2 is below 0.5% for all three coins—equity moves leave almost all of crypto volatility unexplained.
- F-tests sometimes mark coefficients as statistically significant, but the effect size is tiny: a one-standard-deviation equity move shifts predicted crypto vol by only a few basis points.
- Interpretation: large equity moves matter at the margin, but the dominant driver of crypto volatility appears to be crypto-specific forces rather than broad equity conditions.

Asset <chr>	R2_contemp <dbl>	pF_contemp <dbl>	R2_pred <dbl>	pF_pred <dbl>
BTC	0.0040	0.0606	0.0051	0.0279
ETH	0.0026	0.1681	0.0040	0.0613
SOL	0.0043	0.0495	0.0041	0.0570

Limitations

Limitations

- **Sample & events:**
 - Daily data from ~2020–2025, dominated by COVID, the 2021 crypto boom, and the post-2022 tightening cycle. Results may not generalize to calmer periods or other crises.
- **Asset coverage:**
 - Only U.S. large-cap indices (S&P 500, Nasdaq) and three large coins (BTC, ETH, SOL). No small-cap stocks, other regions, stablecoins, DeFi, or sector indices, so the “equity–crypto” linkage is only partially captured.
- **Modeling assumptions (Blocks 2, 3, 4):**
 - Linear VARs with lags chosen by AIC, and DCC-GARCH with relatively simple specifications (Gaussian errors, symmetric dynamics, pairwise systems). Results may be sensitive to lag length, distributional assumptions, and our choice of break dates.
- **Interpretation of “causality”:**
 - Granger tests capture predictive power, not true economic causality. Omitted macro variables (rates, risk appetite, policy news) and structural breaks could drive both equities and crypto, so we cannot claim causal stories.

Conclusions

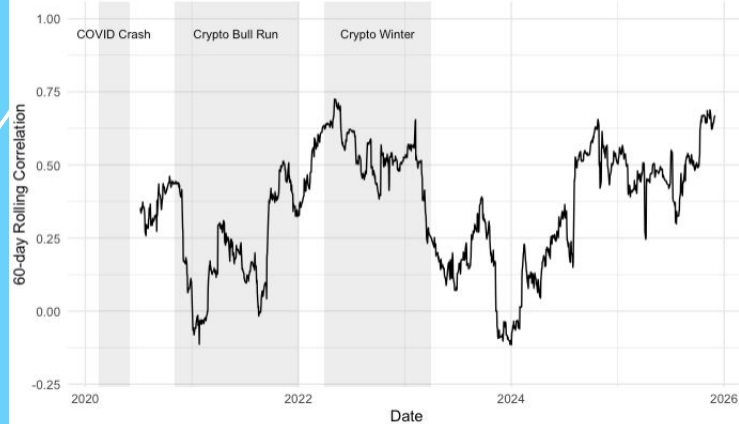
Conclusions

- **Dependence:** Equities and major coins have moderate, time-varying positive correlations, higher with Nasdaq and stronger after 2022.
- **Dynamics:** VAR / Granger and IRFs show weak and short-lived return spillovers in both directions. There exists little true predictability.
- **Volatility:** Equity shocks explain only a tiny share of crypto realized volatility; most vol is crypto-specific.
- **Diversification:** Every 80/20 equity–crypto mix raises long-run returns and Sharpe, but risk and tail are coin and regime dependent.

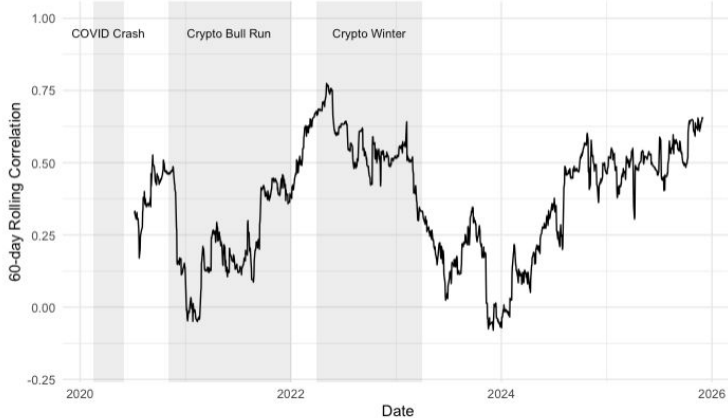
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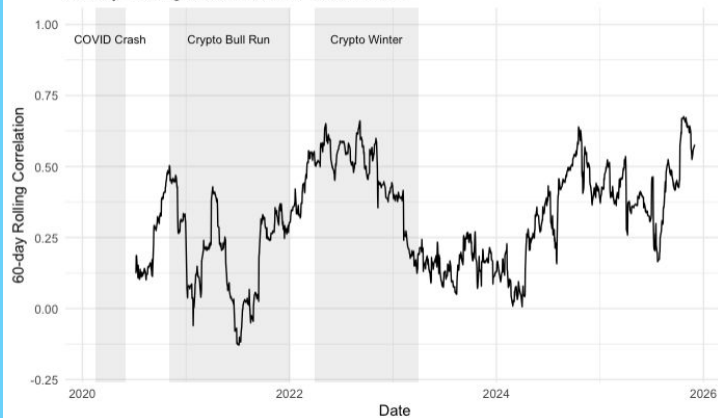
60-day Rolling Correlation: SP500 vs ETH



60-day Rolling Correlation: Nasdaq vs ETH



60-day Rolling Correlation: SP500 vs SOL



60-day Rolling Correlation: ETH vs SOL

